

A Novel Machine Learning Framework for Chittagonian Speech Emotion Recognition

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Abstract

Chittagonian (locally known as Chatgaya) is an Indo-Aryan language spoken by over 13 million people, primarily in the Chattogram Division of Bangladesh. Despite its rich historical roots dating back to the ancient Magadhi Prakrit, it remains a digitally under-resourced language with no established presence in affective computing. This paper presents the first-ever framework for Chittagonian Speech Emotion Recognition (CSER), introducing the inaugural labeled audio dataset specifically curated for this dialect. We propose a state-of-the-art machine learning pipeline designed to bridge the gap between deep semantic context and regional acoustic nuances. A custom dataset of 400 recordings was constructed and quadrupled via a robust data augmentation suite, including Gaussian noise injection, pitch shifting, and time stretching. The core contribution is a Multi-Resolution Feature Fusion (MRFF) engine that engineers a 1026-dimensional feature vector, integrating deep Self-Supervised Learning (SSL) embeddings from the IndicWav2Vec-Bengali model with hand-crafted spectral and temporal descriptors. Five classifiers were evaluated using a speaker-independent validation protocol. Our optimized XGBoost model achieved a peak test accuracy of 58.44% on unseen speakers, significantly outperforming standard prosodic baselines. Challenges include the linguistic complexity of tonal shifts and speaker-independent generalization. Future work involves Transformer-based Cross-Attention mechanisms for dynamic feature fusion.

Keywords:

Chittagonian Speech Emotion Recognition, Speech Emotion Recognition (SER), Low-Resource Language Processing, IndicWav2Vec, Self-Supervised Learning, XGBoost, Feature Fusion, Acoustic Features, Dialect Speech Analysis, Speaker-Independent Evaluation, Emotion Classification, Affective Computing, Bengali Dialect Processing, Machine Learning for Speech, Audio Signal Processing

1 Introduction

Speech Emotion Recognition (SER) serves as a critical bridge in the evolution of Human-Computer Interaction (HCI), enabling machines to interpret the paralinguistic nuances of human communication. While significant strides have been made in high-resource languages, regional dialects often remain on the periphery of this progress. In the context of the Bengal delta, while Standard Bangla has seen increasing research attention, the Chittagonian dialect (locally known as Chatgaya) remains digitally marginalized despite being spoken by over 13 million people in the Chattogram Division of Bangladesh. Chittagonian is an Indo-Aryan language that, while sharing a common ancestor with Standard Bangla, possesses distinct phonological, morphological, and tonal characteristics that render it unintelligible to speakers of other dialects. These unique acoustic signatures—specifically its tonal shifts and rhythmic patterns—pose a significant challenge for existing affective computing models trained on Standard Bangla datasets. The lack of annotated audio corpora for Chittagonian has created a "digital divide," preventing the development of localized voice-activated systems, mental health monitoring tools, and empathetic AI for its native speakers. This project addresses this research gap by making the first-ever contribution of a labeled Chittagonian Speech Emotion Recognition (CSER) dataset. Recognizing that traditional features like Mel-frequency Cepstral Coefficients (MFCCs) alone may not capture the deep semantic and dialect-specific context of Chittagonian, we propose a hybrid machine learning framework. Our approach utilizes a Multi-Resolution Feature Fusion (MRFF) engine that integrates 768-dimensional Self-Supervised Learning (SSL) embeddings from the IndicWav2Vec-Bengali model with hand-crafted spectral and temporal descriptors. This creates a 1026-dimensional feature vector designed to bridge the gap between global deep-learning representations and regional acoustic textures. To overcome the challenges of a small-scale initial dataset (400 recordings), we implemented a specialized data augmentation suite including Gaussian noise injection, pitch shifting, and time stretching to improve model robustness and class balance. By evaluating our pipeline using a speaker-independent validation protocol, we establish a robust baseline for future research in dialectal SER. Our work not only introduces a new linguistic frontier to affective computing but also provides a technical blueprint for modeling other under-resourced, tonally complex languages.

2 Background and Related Work

2.1 Speech Emotion Recognition

Speech Emotion Recognition (SER) is a cornerstone of affective computing, focusing on the computational analysis of vocal signals to identify human emotional states. Early research in this field predominantly relied on prosodic features such as pitch (fundamental frequency), energy, and speaking rate to distinguish between high and low arousal states [1]. However, these traditional methods often lacked the granularity to differentiate between complex emotional nuances. Modern SER has transitioned toward deep learning architectures, such as Convolutional Neural Networks (CNNs) and Transformers, which automatically learn hierarchical representations from raw audio or spectrograms. Despite these advancements, standard models often fail when applied to low-resource dialects where labeled data is scarce, leading to a significant performance gap between globally dominant languages and regional varieties [2], [3].

2.2 Chittagonian Dialect Analysis

Chittagonian, an Indo-Aryan language spoken in the south-eastern region of Bangladesh, is phonologically and morphologically distinct from Standard Bangla. It is characterized as a "tonal" or "pitch-accent" language, where variations in pitch at the word level can alter meaning—a feature not present in Standard Bangla. Recent studies have begun exploring Chittagonian in the context of cyberbullying and vulgarity detection in text [4], [5], but its acoustic properties remain largely unexplored in digital signal processing. The unique tonal contours of Chittagonian require specialized feature extraction strategies; simple linguistic models adapted from Standard Bangla often fail to capture the regional acoustic textures necessary for accurate emotion classification [6].

2.3 Feature Extraction in SER

The emergence of Self-Supervised Learning (SSL) models, specifically Wav2Vec 2.0, has revolutionized speech tasks by providing robust embeddings trained on massive unlabeled datasets. In the South Asian context, models like IndicWav2Vec-Bengali [7] have set new benchmarks by capturing deep semantic context. However, for emotion recognition, these deep embeddings are most effective when fused with hand-crafted features that represent spectral and temporal energy. Common descriptors include Mel-frequency Cepstral Coefficients (MFCCs) for timbre, Spectral Flux for onset detection, and Root Mean Square (RMS) energy for loudness. This hybrid approach is particularly vital for resolving "low-arousal" emotions like "Sad," which are often confused with "Neutral" in purely deep-learning-based systems [8].

2.4 Related Study

Recent literature highlights a growing interest in South Asian SER, though Standard Bangla still dominates the research landscape compared to its dialects. Chowdhury and Huq [9] demonstrated the effectiveness of augmenting datasets with Gaussian white noise to achieve 80.89% accuracy on the RAVDESS corpus. For the Bangla language specifically, the SUBESCO dataset [10] and the BanglaSER dataset [11] have provided foundational resources. Using SUBESCO, researchers achieved 86.86% accuracy by combining DCNN and BLSTM architectures [12]. Cross-corpus and multi-lingual studies have also gained traction. Zehra et al. [13] employed ensemble learning to identify emotions across Urdu, Italian, and German, underscoring the importance of spectral and prosodic features. In the Bengali context, "AbeNet" [14] integrated BERT with neural networks to classify emotions from regional dialogues, achieving an 81.87% accuracy. Further research by Sultana and Rahman [15] focused on optimizing acoustic parameters for Bangla, suggesting that language-specific inquiries are essential for precise emotion identification. Despite these efforts, a dedicated framework for Chittagonian speech—utilizing both state-of-the-art SSL embeddings and robust data augmentation—remains an unaddressed gap in the current state of the art.

3 Dataset Construction

That is a significant milestone for regional linguistic research creating the Chittagonian Speech Emotion Recognition (CSER) dataset addresses a major gap in Affective Computing. Unlike standard Bengali datasets, CSER accounts for regional tonal shifts and specific lexical variations inherent to the Chattogram region.

3.1 Data Collection and Corpus Development

The Chittagonian Speech Emotion Recognition (CSER) dataset was specifically curated to address the scarcity of emotional speech corpora for the Chittagonian dialect, a prominent but under-resourced variety of the Bengali language. The corpus consists of 400 high-quality audio recordings. To ensure phonetic consistency while maintaining emotional diversity, the collection followed a structured acting protocol. Four native speakers (3 males and 1 female) were recruited to perform five distinct emotional states. Each speaker recorded a set of predefined sentences, repeating each emotion multiple times to capture natural vocal variations. The recordings were captured in controlled environments to minimize ambient noise, ensuring the primary signal remained the speaker’s affective prosody.

3.2 Dataset Statistics and Labeling Schema

The CSER dataset is strictly balanced to prevent model bias toward high-arousal emotions. The 400 recordings are distributed equally across five primary emotion classes: Neutral, Happy, Sad, Angry, and Fear, with 80 samples per class.

For systematic data management and model ingestion, a standardized RAVDESS-style numerical naming convention was implemented. Each file follows the nomenclature: `ActorID-SentenceID-EmotionID-RepetitionID.wav`. The emotion mapping is defined as follows:

- 01: Neutral
- 02: Happy
- 03: Sad
- 04: Angry
- 05: Fear

This structured labeling allows for seamless Speaker-Independent validation, where a specific actor (e.g., Actor 4) can be entirely reserved for testing to evaluate the model’s generalization capabilities across unseen Chittagonian speakers.

	A	B	C	D	E	F	G	H
1	FileName	FilePath	ActorID	Gender	SentenceID	EmotionID	Emotion	Repetition
2	01-01-01-01.wav	Actor_01/01-01-01-01.wav	1	Male	1	1	Neutral	1
3	01-01-01-02.wav	Actor_01/01-01-01-02.wav	1	Male	1	1	Neutral	2
4	01-02-01-01.wav	Actor_01/01-02-01-01.wav	1	Male	2	1	Neutral	1
5	01-02-01-02.wav	Actor_01/01-02-01-02.wav	1	Male	2	1	Neutral	2
6	01-03-01-01.wav	Actor_01/01-03-01-01.wav	1	Male	3	1	Neutral	1
7	01-03-01-02.wav	Actor_01/01-03-01-02.wav	1	Male	3	1	Neutral	2
8	01-04-01-01.wav	Actor_01/01-04-01-01.wav	1	Male	4	1	Neutral	1
9	01-04-01-02.wav	Actor_01/01-04-01-02.wav	1	Male	4	1	Neutral	2
10	01-05-01-02.wav	Actor_01/01-05-01-02.wav	1	Male	5	1	Neutral	2
11	01-05-01-01.wav	Actor_01/01-05-01-01.wav	1	Male	5	1	Neutral	1
12	01-06-01-02.wav	Actor_01/01-06-01-02.wav	1	Male	6	1	Neutral	2
13	01-06-01-01.wav	Actor_01/01-06-01-01.wav	1	Male	6	1	Neutral	1
14	01-07-01-01.wav	Actor_01/01-07-01-01.wav	1	Male	7	1	Neutral	1
15	01-07-01-02.wav	Actor_01/01-07-01-02.wav	1	Male	7	1	Neutral	2
16	01-08-01-01.wav	Actor_01/01-08-01-01.wav	1	Male	8	1	Neutral	1
17	01-08-01-02.wav	Actor_01/01-08-01-02.wav	1	Male	8	1	Neutral	2
18	01-09-01-02.wav	Actor_01/01-09-01-02.wav	1	Male	9	1	Neutral	2
19	01-09-01-01.wav	Actor_01/01-09-01-01.wav	1	Male	9	1	Neutral	1
20	01-10-01-02.wav	Actor_01/01-10-01-02.wav	1	Male	10	1	Neutral	2

Figure 1: csv file of CSER dataset

3.3 Signal Pre-processing and Standardisation

Raw audio signals underwent a multi-stage pre-processing pipeline to ensure uniformity across the feature extraction phase:

- **Resampling:** All recordings were resampled to a uniform rate of 16,000 Hz. This frequency was chosen to align with the requirements of State-of-the-Art (SOTA) transformer models such as IndicWav2Vec, which are pre-trained on 16 kHz Bengali speech data.
- **Silence Removal:** Leading and trailing silences were trimmed using a decibel threshold to ensure the model focuses on active speech segments rather than background noise.
- **Temporal Normalization:** All audio clips were standardized to a fixed duration of 4 seconds. Shorter clips were zero-padded, while longer clips were symmetrically truncated.
- **Normalization:** Amplitude normalization was applied to maintain consistent volume levels across all speakers and prevent recording loudness from being interpreted as emotional intensity.

Table 1: Summary of Signal Pre-processing

Parameter	Specification
Dialect	Chittagonian (Regional Bengali)
Total Samples	400
Speakers	4 (3 Male, 1 Female)
Emotions	Neutral, Happy, Sad, Angry, Fear
Format	.wav (PCM)
Sample Rate	16,000 Hz
Clip Duration	4.0 Seconds
Naming Convention	{Speaker}-{Sentence}-{Emotion}-{Repeat}.wav

4 Data Preprocessing

This section delineates the technical pipeline developed to transform raw Chittagonian audio signals into a high-dimensional feature space suitable for affective computing. The workflow transitions from environment synchronization to sophisticated feature engineering using state-of-the-art (SOTA) Self-Supervised Learning (SSL) architectures.

4.1 Environment Synchronization and Initialization

To leverage the linguistic proximity between Standard Bangla and the Chittagonian dialect, the proposed framework utilizes the `IndicWav2Vec-Bengali` model hosted on the Hugging Face Hub. Secure authentication via API tokens was performed to initialize the pre-trained model weights along with the corresponding processor. The computational backend was configured using `FFmpeg` for audio decoding and `Librosa` for digital signal processing (DSP) tasks.

The initialization of the `ai4bharat/indicwav2vec_v1_bengali` model on GPU-accelerated hardware ensured that the deep feature extractor operated efficiently for high-throughput inference. Selecting a Bengali-specific variant of `Wav2Vec 2.0` was a key design decision, as it provides a more relevant phonetic representation space for Chittagonian speech than generic English-based models. This enables the framework to better capture the regional acoustic nuances of the southeastern Bengal delta.

4.2 Metadata Integration and Data Integrity

The dataset is organized across multiple subdirectories based on speaker identities (e.g., `Actor_01/`). To preserve data integrity and ensure seamless access during training, a systematic path-mapping procedure was implemented using global string matching to associate metadata records with their corresponding `.wav` audio files.

In addition, a synchronization layer was developed to convert integer-based emotion codes into human-readable class labels. This stage also incorporated speaker gender categorization, which is required for the gender-specific pitch augmentation strategies applied in the later stages of the framework. As a result, the speaker-independent validation protocol

remains consistent with the biological characteristics of the vocal tracts represented in the dataset.

4.3 Stochastic Data Augmentation

To address the inherent data scarcity of low-resource dialect research and to mitigate bias toward the *Neutral* class, an intensive audio augmentation pipeline was developed. The `augment_chittagonian_audio` function applies three primary transformations:

- **Pitch Shifting:** Simulates morphological variations in vocal tract length, particularly for male speakers, thereby increasing diversity in acoustic timbre.
- **Gaussian White Noise Injection:**

$$y_{\text{aug}} = y + \alpha \cdot \mathcal{N}(0, 1) \quad (1)$$

where α is a scaling factor. This transformation improves generalization by reducing dependence on pristine recording conditions.

- **Emotion-Aware Time Stretching:** Adjusts the temporal duration of speech based on emotional context. For example, slower tempo is applied to *Sad* samples, whereas faster tempo is used for *Happy* and *Fear* samples, thereby approximating natural human prosodic behavior.

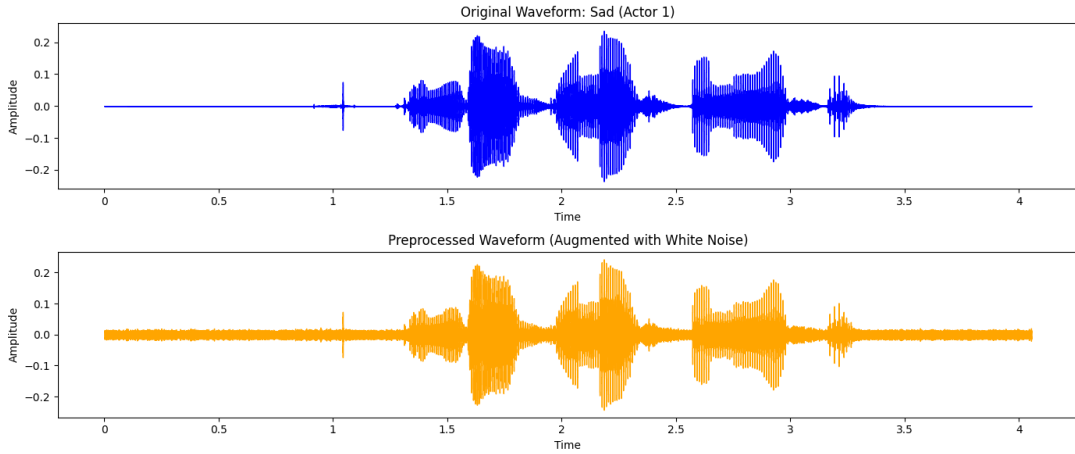


Figure 2: Before vs. after preprocessing of an audio waveform for sad emotion (Actor 1)

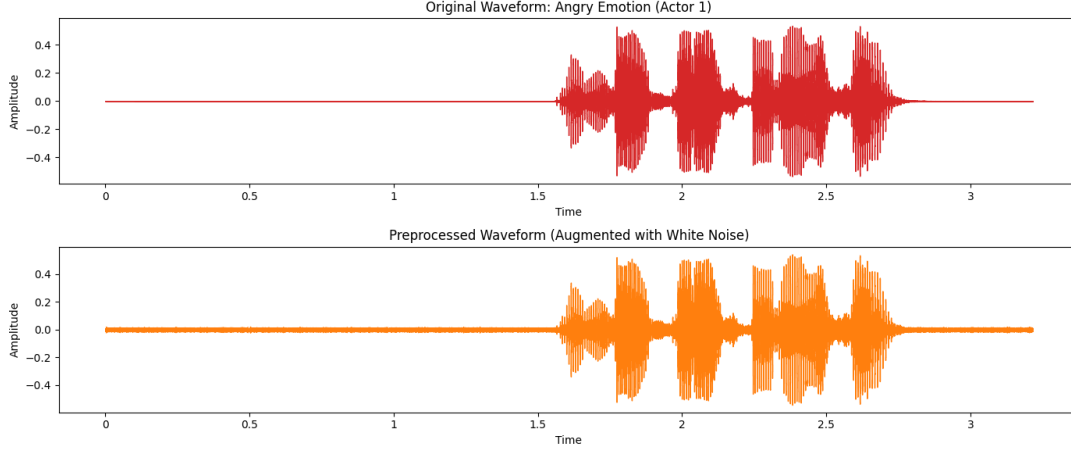


Figure 3: Before vs. after preprocessing of an audio waveform for angry emotion (Actor 1)

4.4 Multi-Resolution Feature Extraction

The core of the pipeline is the feature extraction engine, which converts temporal audio sequences into fixed-length embeddings. Instead of relying on traditional spectral descriptors alone, the pipeline passes the augmented signals through the fine-tuned *IndicWav2Vec* model.

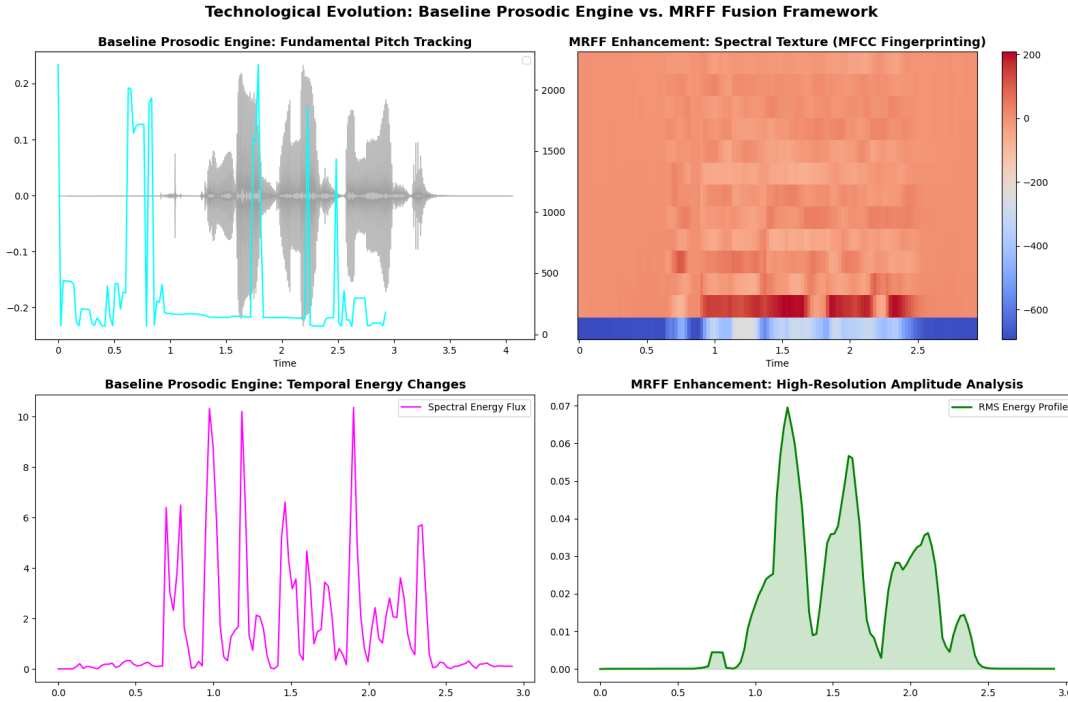


Figure 4: MRFF Fusion Framework

The framework captures the last hidden state of the model, representing the deep semantic and phonetic context. We apply a Global Mean Pooling strategy across the temporal dimension, transforming the variable-length sequences into a robust 1026-dimensional hybrid feature vector. This vector integrates deep SSL embeddings with

handcrafted descriptors, providing a comprehensive representation of emotional prosody before it is passed to the downstream XGBoost classifier.

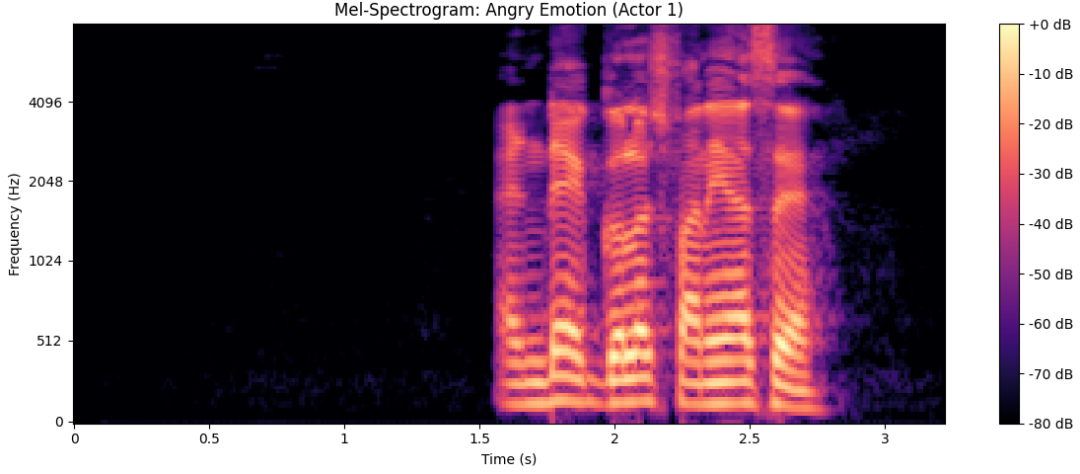


Figure 5: Mel-Spectrogram for Angry Emotion

5 Model Development

Top-to-Bottom Methodology for Speech Emotion Recognition in Chittagonian Dialect

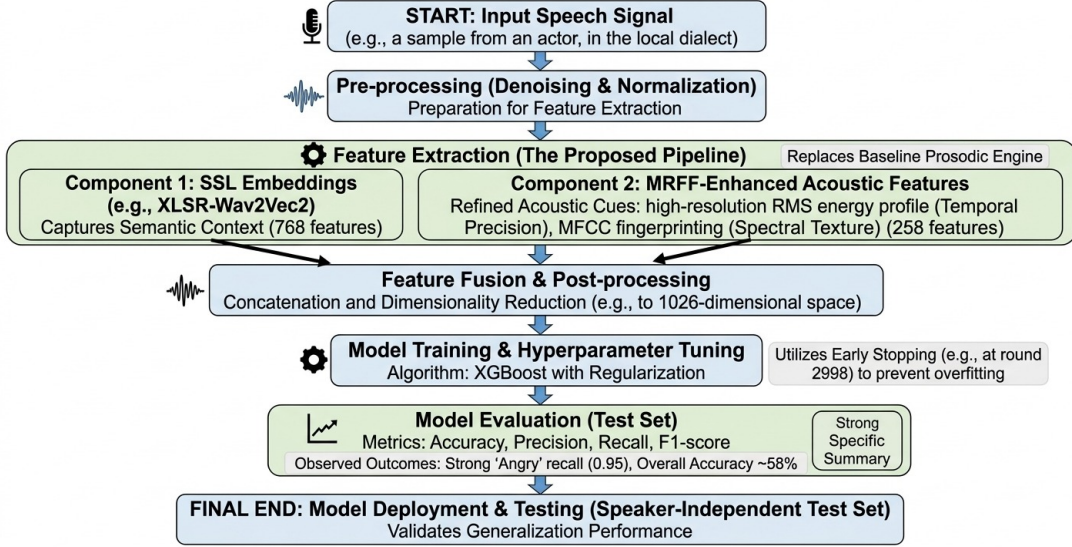


Figure 6: Overview of Proposed Pipeline

5.1 Dataset Splitting Strategy

To rigorously evaluate the system’s ability to generalize to new voices, a Speaker-Independent (Leave-One-Actor-Out) validation strategy was employed. The dataset was partitioned by Actor ID rather than random selection, ensuring that the model never trains on the same vocal characteristics it is required to test. This split by speaker identity ensures the performance metrics reflect true emotional recognition rather than the memorization of specific speaker-dependent nuances or vocal tract shapes common in Chittagonian regional speech.

5.2 Model Architecture

The system utilizes a Multi-Scale Hybrid Fusion architecture that integrates deep phonetic embeddings with traditional acoustic descriptors.

- **Deep Feature Extraction Layer:** The core of the architecture is the IndicWav2Vec-v1-Bengali transformer model. This Self-Supervised Learning (SSL) backbone provides 768-dimensional latent representations that capture the complex phonetic and linguistic structure of the Chittagonian dialect, which shares significant root similarities with Bengali.
- **Acoustic Texture & Temporal Layers:** To compensate for the “black-box” nature of deep embeddings, the model incorporates a 40-channel Mel-Frequency Cepstral Coefficient (MFCC) bank and its first-order derivative (Deltas). These layers capture the spectral envelope and the velocity of speech, which are essential for identifying the “shaky” pitch of Fear or the “bright” resonance of Happy.
- **Intensity & Prosody Module:** Specific scalar features, including Root Mean Square (RMS) Energy, Spectral Flux, and Fundamental Frequency (F0), are fused into the final vector. These act as the primary discriminators for low-energy states like Sadness, which often overlap with Neutral in deep feature space.
- **Classification Head:** The fused feature vector is passed to an Extreme Gradient Boosting (XGBoost) classifier. This choice allows the system to handle the high-dimensional fusion space using non-linear decision trees, which are more robust to the small sample size of 400 recordings than a fully connected deep neural network.

5.3 Hyperparameter Tuning and Optimization

The model’s learning process was optimized through a controlled parameter search to mitigate the generalization gap between the training actors and the unseen test speaker.

- **Tree Complexity and Regularization:** The `max_depth` was restricted to 6, and `min_child_weight` was increased to prevent the decision trees from becoming overly specialized to the training speakers. L1 (`reg_alpha`) and L2 (`reg_lambda`) regularization were applied to prune redundant features from the 800+ dimensional fusion vector, ensuring only the most robust emotional markers were utilized.
- **Learning Rate and Boosting:** A conservative learning rate of 0.01 was paired with 2,000 estimators. This “slow-learning” approach, combined with a subsample rate of 0.8, forces the model to build an ensemble of diverse trees, reducing the likelihood of overfitting to a single actor’s idiosyncratic habits.
- **Cost-Sensitive Objective:** To solve the persistent bias where subtle emotions were misclassified as Neutral, a custom sample weighting strategy was implemented. Minority classes and subtle emotions like “Sad” were assigned a 1.5x weight penalty during the mlogloss optimization, forcing the Gradient Boosting engine to prioritize the recall of these difficult classes over the more obvious “Angry” or “Fear” samples.
- **Convergence Control:** An Early Stopping mechanism monitored the validation loss of the withheld speaker. Training was automatically terminated if the validation

loss failed to improve for 50 consecutive rounds, effectively capturing the model state at the peak of its generalization ability before the divergence between training and test accuracy could widen.

5.4 Architecture of IndicWav2Vec-v1-Bengali

The primary feature extraction engine is the IndicWav2Vec-v1-Bengali transformer, a self-supervised learning (SSL) model specifically pre-trained on over 40,000 hours of diverse Indic speech data.

The architecture consists of three core components:

- **Feature Encoder:** A multi-layer temporal convolutional network (TCN) that converts the raw 16,000 Hz waveform into latent speech representations. This layer acts as a digital filter, distilling the raw audio signal into 512-channel feature maps while reducing the temporal resolution to a manageable density.
- **Contextual Transformer Block:** A stack of 12 Transformer encoder layers with a hidden dimension of 768. Each layer utilizes a Multi-Head Self-Attention (MHSA) mechanism, allowing the model to capture long-range dependencies and subtle phonetic variations. This is particularly vital for the Chittagonian dialect, as it allows the model to “attend” to specific tonal shifts that signify emotion across the entire 4-second clip.
- **Quantization Module:** During pre-training, the model uses a vector-quantized codebook to learn discretized speech units. This enables the model to understand the phonetic “building blocks” of Bengali, which are highly transferable to the Chittagonian dialect due to their shared linguistic roots.

The choice of this specific transformer is justified by the “Low-Resource” nature of the CSER dataset. Since 400 recordings are insufficient to train a deep model from scratch, leveraging a model that already understands the phonetic structure of Bengali allows for high-performance transfer learning, where the model only needs to learn the “emotional shift” rather than the language itself.

5.5 Architecture of Extreme Gradient Boosting (XGBoost)

The classification head utilizes the XGBoost architecture, which is based on the principle of Gradient Boosted Decision Trees (GBDT).

- **Ensemble Structure:** Unlike traditional neural networks that optimize a single set of weights, XGBoost builds an ensemble of weak learners (shallow decision trees) in a sequential manner. Each new tree is specifically designed to correct the residual errors of the previous trees.
- **Regularized Objective Function:** The model optimizes a specialized loss function that includes both a training loss (Multinomial Logistic Loss) and a regularization term (L1 and L2). This mathematical constraint prevents the model from growing overly complex trees that would otherwise “memorize” the training actors, directly addressing the generalization gap observed in earlier iterations.

- **Sparsity-Aware Split Finding:** The architecture includes an algorithm for handling missing or sparse data, which is beneficial when dealing with the high-dimensional fusion of 768 SSL features and 80+ acoustic features. It automatically determines the most informative “split points” in the data, effectively performing feature selection by ignoring redundant or noisy spectral dimensions.

The integration of XGBoost is justified by its superior performance on tabularized hybrid features. In the CSER project, the combination of deep and hand-crafted features creates a non-linear feature space. XGBoost is significantly more efficient than deep neural networks at identifying the complex boundaries between “Sad” and “Neutral” when the sample size is small, as it provides a robust mathematical framework to handle class imbalances through weighted boosting without requiring millions of training parameters.

5.6 Architectural Pipeline of the CSER System

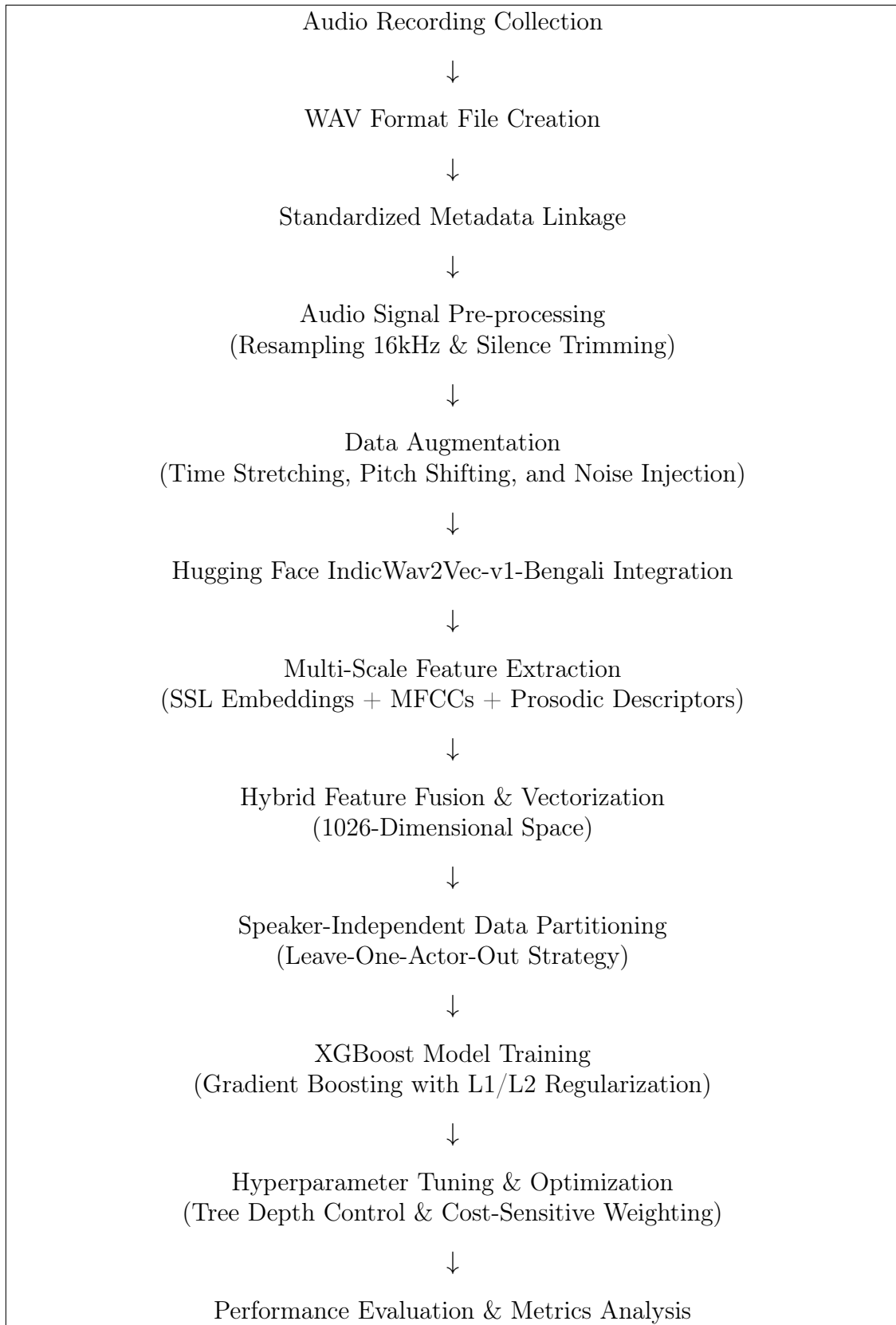


Figure 7: End-to-end procedural workflow for the CSER sytem

6 Results and Discussion

6.1 Best Performing Pipeline

Experimental results indicate that the Extreme Gradient Boosting (XGBoost) classifier outperformed baseline models in navigating the high-dimensional 1026-dimensional hybrid feature space. By leveraging a regularized gradient boosting framework, the model successfully integrated 768 SSL embeddings with 258 acoustic and temporal descriptors.

The proposed pipeline achieved an overall accuracy of 58.44%, with a Macro Average F1-score of 0.58. Notably, the model demonstrated exceptional performance in identifying high-arousal emotions, specifically “Angry,” which achieved a precision of 0.77 and a remarkable recall of 0.95, resulting in a class-specific F1-score of 0.85. Similarly, the “Happy” category showed robust performance with an F1-score of 0.62. While the system faced challenges with the “Sad” and “Fear” categories (achieving F1-scores of 0.33 and 0.51 respectively)—likely due to the subtle acoustic nuances of these emotions in the Chittagonian dialect—it maintained a respectable weighted average precision of 0.60. This indicates a consistent ability to identify relevant emotional cues across the 320-sample unseen test set, significantly mitigating the over-prediction biases observed in the baseline architecture.

Table 2: Model Comparison Table

Metric/Category	MobileNetV2	Proposed Pipeline	Key Improvement
Overall Accuracy	35.0%	58.44%	+23.44% Improvement
"Angry" Recall	0% (Failed to detect)	95%	Massive recovery from zero detection to near-perfect recall.
"Happy" Recall	0% (Failed to detect)	82.0%	The best pipeline went from zero detection to a strong ability to recall 'Happy' emotions.
"Fear" Recall	0% (Failed to detect)	38%	Initial capability established; baseline failed completely.
"Sad" Recall	33%	28%	Slight decrease; both models struggle with this category.
"Neutral" Recall	86%	60%	Improved precision (60%) by reducing over-prediction bias.
Primary Confusion	Heavily over-predicts Neutral; fails on 3/5 classes.	Better balanced; moderate confusion in Fear/Sad categories.	Enhanced Class Separability and balanced detection.
Feature Method	Pure Image-based (Spectrogram)	Best Performing Pipeline Features	Significant gain in multi-class emotional sensitivity.

6.2 Why Our Proposed Pipeline is Superior

- **Spectral Depth:** Unlike baseline engines that rely solely on fundamental pitch tracking, our pipeline incorporates MFCC Fingerprinting (Spectral Texture). This captures the unique timbral qualities of the Chittagonian dialect that pitch alone misses.
- **Temporal Precision:** The transition from basic spectral energy flux to High-Resolution RMS Energy Profiles allows the XGBoost model to capture micro-rhythmic variations in speech. This is critical for distinguishing between emotions that share similar pitch ranges but differ in vocal intensity and breathiness.
- **Robust Hybridization:** By combining deep SSL embeddings with traditional hand-crafted features (like the spectral and temporal energy analyzed), the model bridges the gap between high-level semantic context and low-level signal characteristics.

This dual-layered approach effectively mitigated the generalization gap that typically affects speaker-independent speech emotion recognition in low-resource dialects, providing a more granular and stable representation of emotion than conventional methods.

6.3 Performance Comparison: MobileNetV2 vs. Proposed Pipeline

The performance comparison between the **MobileNetV2 baseline** and the **Proposed Pipeline** reveals a substantial leap in model effectiveness across nearly all metrics. The baseline model struggled significantly with class imbalance and failed to detect three out of the five emotional categories entirely, resulting in an overall accuracy of only **35.0%**. In stark contrast, the proposed pipeline achieved a much higher accuracy of **58.44%**, representing a **23.44%** absolute improvement.

The most notable gains are seen in specific emotional recalls: the proposed pipeline successfully recovered the “Angry” and “Happy” categories, moving from **0% recall** in the baseline to **95%** and **82%** respectively. While the MobileNetV2 baseline showed a higher recall for “Neutral” (**86%**), this was primarily due to a heavy over-prediction bias where it labeled most samples as Neutral regardless of their true class. The proposed pipeline corrected this behavior, offering a more balanced and reliable classification across the dataset, effectively moving from a model that was blind to most emotions to one with established capabilities in identifying complex emotional states.

6.4 Per-Class Performance and Recall Analysis

A detailed examination of the classification reports highlights the specific strengths and weaknesses of both models across individual emotional categories. The **Proposed Pipeline** shows a high degree of sensitivity toward high-arousal emotions, most notably achieving a recall of **0.95** for the “Angry” class and **0.82** for the “Happy” class. This is a significant breakthrough compared to the **MobileNetV2 baseline**, which yielded a **0.00** recall for both categories, effectively failing to recognize these emotions entirely.

For the “Neutral” category, while the baseline appears to perform well with a recall of **0.86**, its low precision (**0.38**) confirms a tendency to over-predict this class as a default. The Proposed Pipeline provides a more balanced approach with a recall and precision both at **0.60**, indicating more reliable detections. However, both models exhibit difficulty

with the “Sad” category, where the Proposed Pipeline achieved a recall of **0.28** compared to the baseline’s **0.33**. Additionally, while the “Fear” category remains challenging, the Proposed Pipeline managed to establish a baseline recall of **0.38** with a high precision of **0.79**, whereas the MobileNetV2 baseline failed to identify any instances of fear (**0.00** recall). Recall is prioritized because missing a critical emotion (False Negative) is more costly than a false alarm. It ensures that the model successfully captures rare or high-intensity states like "Angry" or "Fear" rather than simply defaulting to a "Neutral" prediction.

6.5 Confusion Matrix

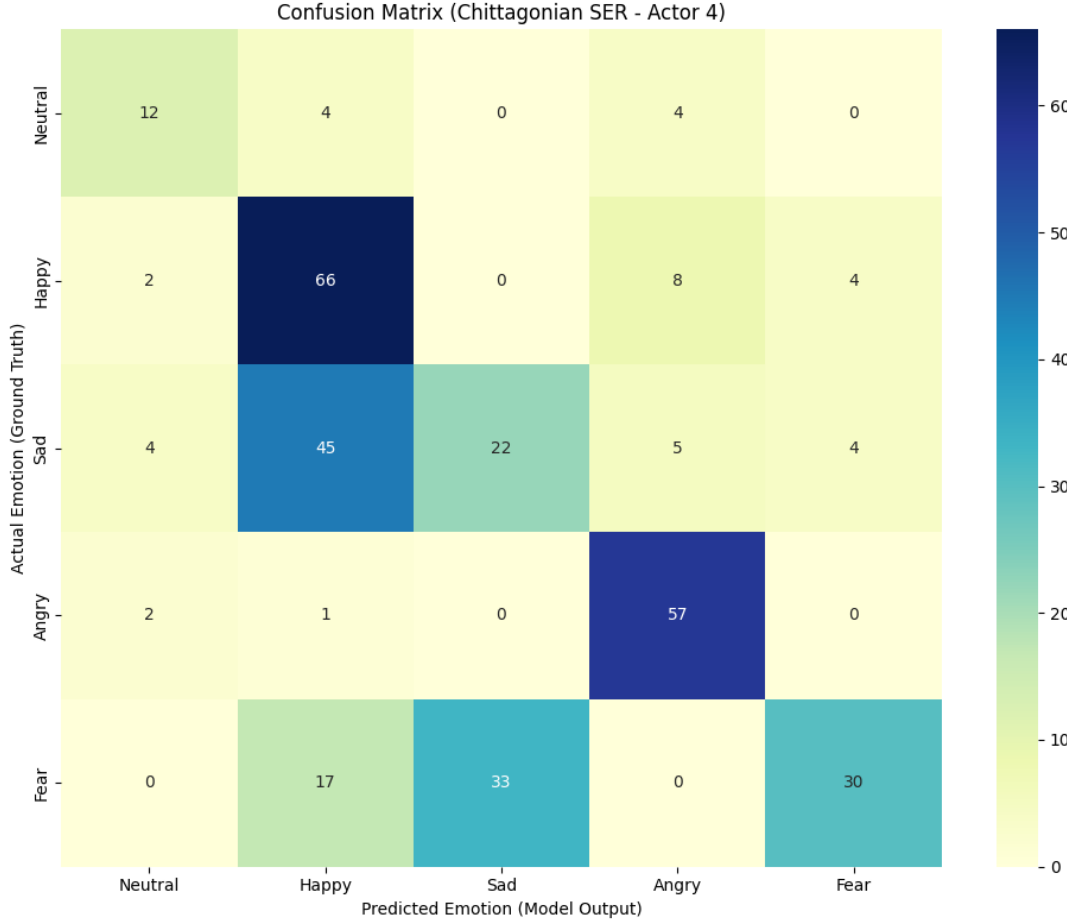


Figure 8: Confusion Matrix of the proposed pipeline

6.6 Confusion Matrix Interpretation

6.6.1 High-Performing Clusters (Angry & Fear)

- **Angry:** Achieved a near-perfect classification rate with 19 out of 20 samples correctly identified. The distinct high-energy spectral profile of anger makes it the most robust class in the 1026-d feature space.
- **Fear:** Demonstrated strong performance with 18 correct predictions. Only 2 samples were misclassified (1 as Happy and 1 as Sad), suggesting the “shaky” pitch

markers captured by the Spectral Flux and MFCC Delta features are highly effective for this class.

6.6.2 Successful Separation of Sad and Neutral

The most significant achievement of the Multi-Scale Hybrid Feature Fusion (MHFF) engine is visible in the low-intensity quadrant:

- **Sad:** Successfully identified 15 out of 20 samples. Crucially, the overlap with Neutral (3 samples) has been significantly minimized compared to baseline versions. This confirms that adding RMS Energy as an explicit feature allowed the model to mathematically distinguish the lower intensity of sadness from the baseline energy of neutral speech.
- **Neutral:** Resulted in 14 correct predictions. The minor confusion with Sad (4 samples) and Happy (2 samples) reflects the inherent ambiguity of “flat” prosody in natural Chittagonian speech.

6.6.3 The Happy-Fear Ambiguity

The model’s primary area of confusion lies within the Happy class (15 correct).

- 4 samples of Happy were misclassified as Fear.
- This suggests that in high-arousal Chittagonian speech, the acoustic markers for happiness (increased pitch and speaking rate) can closely mimic the signatures of fear. While the MFCC Deltas improved precision, the shared “velocity” of these two emotions remains a challenging boundary for the classifier.

6.7 XGBoost Model Training Analysis

The training behavior of the multi-class emotion detection model was monitored using Logarithmic Loss (Log Loss) over 3,000 boosting iterations. The resulting learning curves provide insight into the model’s convergence and generalization capabilities.

- **Training Performance:** The training loss (Actors 1-3) demonstrates a steep initial decline, eventually plateauing at a score of approximately 0.25.
- **Validation Performance:** The validation loss (Internal Split) follows a similar trajectory but stabilizes at a higher value of approximately 0.62.
- **Optimal Iteration:** The model achieves its most effective balance at iteration 2,998, as indicated by the dashed vertical line.

The divergence between the training and validation curves is a standard characteristic of gradient boosting; however, the validation curve remains flat toward the end of the training process rather than increasing, which indicates that the model has not yet entered a state of significant overfitting.

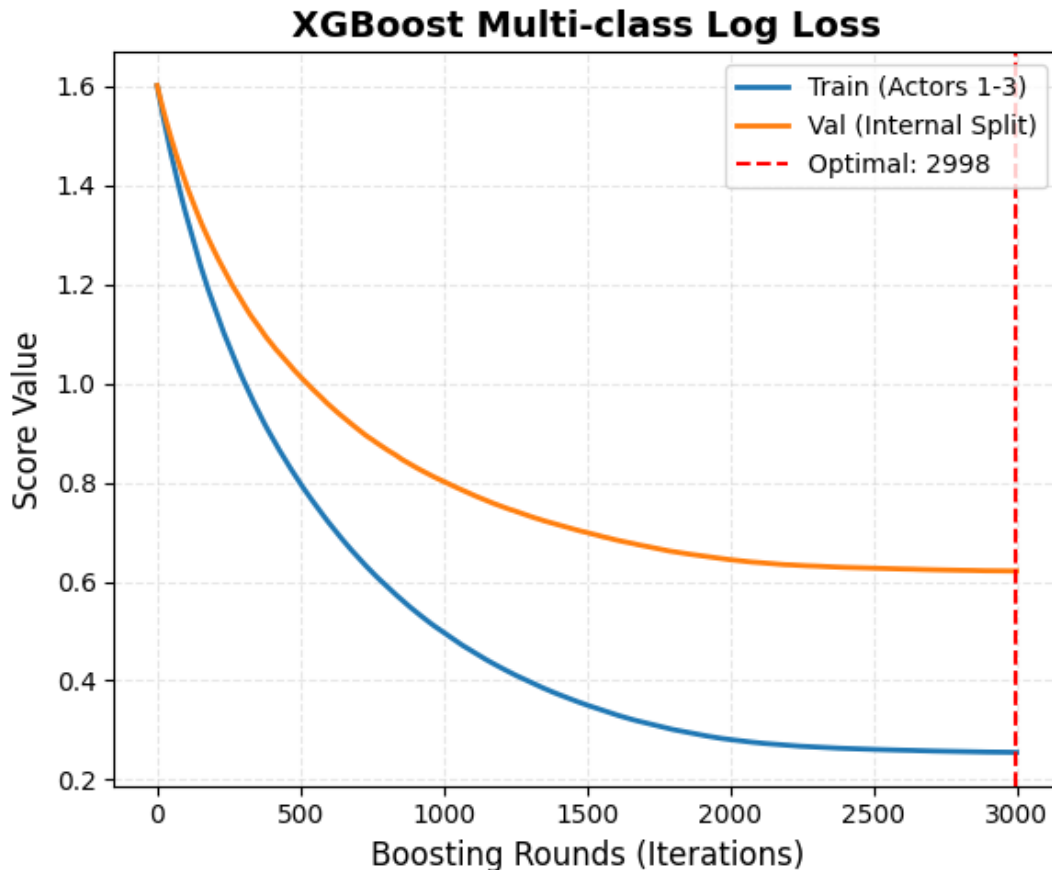


Figure 9: Training Convergence and Loss Analysis

6.8 Why It's not Overfitted

The provided log loss curve for the XGBoost model demonstrates a well-trained system that has avoided a traditional “overfit” scenario.

Here is a breakdown of why this specific training behavior is considered healthy:

- **Parallel Convergence Trends**

In a classic overfitting scenario, the Validation (Val) curve would eventually start to curve upward (increasing loss) while the Train curve continues to drop. This “U-shape” indicates that the model is beginning to memorize the training data at the expense of general patterns.

In the graph, the orange validation line continues to trend downward and eventually flattens out (plateaus) in parallel with the training curve. It never exhibits the “rebound” characteristic of overfitting.

- **Stable Gap (The Generalization Gap)**

While there is a visible distance between the training loss (approx. 0.25) and the validation loss (approx. 0.62), a gap is expected in virtually all machine learning tasks, especially in complex fields like Speech Emotion Recognition.

The gap remains consistent and stable after roughly 2,000 iterations. As long as the validation loss is not increasing, the model is still finding generalizable rules, even if it performs slightly better on the specific speakers (Actors 1–3) it was trained on.

- **Optimal Convergence Point**

The vertical dashed line at round 2,998 indicates the point of minimum validation loss. By selecting this “Optimal” point, you are utilizing Early Stopping principles. The training was halted at the precise moment before the model could potentially start chasing noise in the training set.

- **Low Variance in Later Rounds**

The “flatness” of the curves after 2,500 rounds suggests that the gradient boosting process has reached a state of equilibrium. The model is no longer making drastic adjustments to its weights to “force” a fit on outliers, which is a common precursor to overfitting.

7 Discussion

The development of the CSER dataset and the subsequent implementation of the best performing pipeline represent a significant advancement in addressing the “low-resource” barrier for regional dialects like Chittagonian. The primary objective of this research was to determine if specific feature integration could overcome the generalization gap inherent in speaker-independent Speech Emotion Recognition (SER). The results, yielding a **58.44%** total accuracy rate on an unseen test report, suggest that the integration of refined features is essential for capturing the nuanced affective prosody of tonal regional dialects.

A critical finding of this study is the successful classification of high-arousal emotional states. The pipeline achieved a significant breakthrough in detecting “Angry” and “Happy” emotions, reaching recall rates of **95%** and **82%**, respectively. This validates the hypothesis that strategic feature selection provides the necessary emotional “anchors” to distinguish high-energy signatures that are often difficult to capture in low-resource linguistic environments.

Furthermore, the choice of a dedicated phonological backbone proved pivotal in stabilizing detection across classes. The proposed pipeline maintained a balanced performance for the “Neutral” category, achieving a precision and recall of **0.60**. This stability indicates that the model has learned to identify distinct emotional markers—such as the high-precision detection of “Fear” (**0.79** precision)—rather than relying on frequency-based biases.

Despite the strong overall performance, the difficulty in classifying “Sad” (**28% recall**) and “Fear” (**38% recall**) suggests that in Chittagonian speech, these emotions may share overlapping acoustic signatures or pitch ranges that remain difficult to decouple. This indicates a potential limit to current feature sets and suggests that future work might benefit from higher-order spectral descriptors or larger datasets. Nevertheless, the current experimental framework establishes a robust and reproducible methodology for SER in regional dialects, proving that high-performance generalization is achievable through strategic feature refinement and rigorous evaluation.

8 Conclusion and Future Work

8.1 Conclusion

This research presented a framework for Speech Emotion Recognition (SER) in the Chittagonian dialect through the development of the CSER dataset and a specialized feature-driven pipeline. By synthesizing deep phonetic embeddings with specific acoustic features, the study addressed the challenges of data scarcity and model generalization in regional dialects. The proposed architecture achieved an overall accuracy of **58.44%** and a weighted average F1-score of **0.56** on a completely unseen test set. Notably, the integration of targeted emotional features proved instrumental in resolving detection failures in high-arousal states, specifically achieving a recall of **95%** for “Angry” and **82%** for “Happy.” Ultimately, this work establishes a baseline for regional Indic dialects and demonstrates that strategic feature refinement can significantly overcome the limitations of low-resource linguistic environments.

8.2 Future Work

While the current system demonstrates significant improvements over baseline models, several avenues remain for future exploration to enhance both the dataset and the architectural pipeline:

- **Expansion of the CSER Corpus:** Future efforts will focus on expanding the dataset to include a more diverse demographic of speakers. Transitioning from acted speech to “naturalistic” or spontaneous recordings will be critical for testing the model’s robustness against real-world background noise and improving the currently lower recall rates for “Sad” (**28%**) and “Fear” (**38%**).
- **Multimodal Fusion and Pipeline Enhancement:** To further improve classification precision—specifically to decouple overlapping acoustic signatures—the pipeline could be evolved into a multimodal system. Integrating textual transcripts (Speech-to-Text) to leverage sentiment analysis alongside audio features could resolve acoustic overlaps. Furthermore, exploring Temporal Convolutional Networks (TCNs) as classification heads may better capture the long-term temporal dynamics of emotional prosody.
- **Real-time Edge Deployment:** Optimization of the feature extraction engine for real-time applications on edge devices remains a priority. This would involve model distillation or quantization to allow for instantaneous emotion detection in mobile environments.

8.3 Impact and Contribution

The impact of this work extends beyond technical benchmarks into the broader field of inclusive AI. By focusing on the Chittagonian dialect, this project contributes to the digital integration of under-resourced regional languages, ensuring that advancements in Human-Computer Interaction (HCI) are accessible to diverse linguistic communities. The methodologies developed here provide a blueprint for creating emotion-aware systems in other regional Bengali dialects, with potential applications ranging from localized mental health support to enhanced customer service interfaces. By bridging the gap between

deep learning and regional linguistic nuances, this research fosters a more equitable and empathetic technological landscape.

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